

SV-Detect: AI-generated Text Detection with Steering Vectors

Mikhail Vishnyakov
Independent Researcher
mikhail.n.vishnyakov@gmail.com

Tatiana Gaintseva
Queen Mary University of London
t.gaintseva@qmul.ac.uk

Abstract

Detecting AI-generated text is especially difficult under distribution shift, such as transfer across domains, source models, and editing attacks. We propose an AI-generated text detector based on *steering vectors* extracted from the hidden representations of a frozen language model. At each layer, we construct a direction that separates human-written from AI-generated text, and represent each input by its layer-wise alignment with these directions. A lightweight classifier trained on these projection features yields the final detection score. Our method achieves strong performance both in-distribution and under distribution shift, including across domains, source models, and machine-editing transformations such as polishing and rewriting. Interpretation analyses show that the learned directions align with recognizable stylistic cues while capturing substantial additional signal beyond surface features. These results position AI-generated text detection as a representation-space probing problem and show that steering vectors provide a simple and effective solution. Code is available at <https://github.com/Atmyre/sv-detect/>

1 Introduction

The rapid deployment of large language models has made AI-generated text increasingly fluent, diverse, and difficult to distinguish from human writing. (Wu et al., 2024, 2025). This creates practical challenges for content moderation, authorship verification, and benchmark integrity, and has led to growing interest in methods for AI-generated text detection (Wu et al., 2025; Weber-Wulff et al., 2023). At the same time, the problem has become harder: modern detectors must operate not only on direct generations, but also under distribution shift, including transfer across domains, source models, and editing-based attacks such as paraphrasing, polishing, or rewriting (Kehkashan et al., 2025).

• **Predicted: AI-generated** (ground truth: AI-generated)
The negatives of the 69 photographs, which were taken by Greater Manchester Fire and Rescue Service, were recently found in their archives. They show the effects of a bygone era when industrial landscapes and urban challenges merged with acts of courage. Each image serves as a silent testament to dramatic moments recorded on film—the billowing smoke against twilight skies, the...

• **Predicted: Human** (ground truth: Human)
The cash will help give lowest-earning households cheaper bills, replace ageing pipes and restore Cardiff and Lisvane reservoirs. The not-for-profit company, which serves people in Wales and Herefordshire, invested an extra £32 million last year. The plan follows a consultation with 12,000 customers. In total, £5m will go to providing cheaper bills for low...

Figure 1: Token-level steering vector projections distinguish LLM-generated text (top) from human writing (bottom). Word saturation reflects each token’s signed contribution to the classifier.

A large body of prior work approaches this problem through token-level statistics, likelihood-based scores, perturbation tests (Bao et al., 2023; Yang et al., 2023; Hans et al., 2024), or supervised classifiers trained directly on text (Jawahar et al., 2020; Chen et al., 2024). While these methods can be effective in-distribution, they often degrade under transfer, suggesting that many detectors rely on artifacts tied to a particular generator, dataset, or attack style (Li et al., 2024; Wu et al., 2024; Zha et al., 2025). This motivates a different perspective: instead of using solely surface form or output probabilities, can we identify a more stable signal in the *intermediate representations* of a frozen LLM?

In this paper, we propose **SV-Detect**, an AI-generated text detector based on *steering vectors* — AI-generated text directions extracted from the hidden states of a frozen language model. Our central hypothesis is that **human-written and AI-generated texts induce systematically different directions in representation space**, and these directions can be used as robust detection features. Concretely, for each transformer layer, we construct a vector that separates human and AI-generated texts, represent a new input by its layer-

wise alignment with these directions, and train a lightweight classifier on the resulting projection features. This yields a simple detector that does not require fine-tuning the underlying language model and naturally supports interpretation at the level of layers and directions.

We evaluate SV-Detect on several complementary benchmarks. Our main evaluation covers *DetectRL* (Wu et al., 2024) and *MIRAGE* (Fu et al., 2025): *DetectRL* tests robustness across domains, source models, and attack families, and *MIRAGE* targets direct generation and machine-assisted editing scenarios, including polishing and rewriting. We further assess out-of-distribution generalization by transferring detectors trained on *DetectRL* and *MIRAGE* across benchmarks, including transfer between the two and transfer to *PAD-Ben* (Zha et al., 2025). Across these settings, SV-Detect achieves near-perfect in-distribution performance, robust transfer, and strong cross-benchmark generalization. It also substantially outperforms ReprGuard (Chen et al., 2025), the closest representation-based detector, under matched-backbone comparisons.

Beyond accuracy, we analyze what SV-Detect is actually using. The learned steering directions align with interpretable lexical and stylistic cues, but they also capture substantial signal beyond hand-picked surface features. This suggests that SV-Detect is not only effective, but also a useful tool for studying how machine-generated text differs from human writing at the representation level. Overall, our contributions are as follows:

- We **introduce SV-Detect**, a simple and effective AI-generated text detector based on directions extracted from the hidden representations of a frozen language model.
- We **demonstrate strong generalization** across a set of benchmarks, including cross-domain, cross-source, cross-attack and cross-benchmark transfer.
- We **provide interpretation analyses** showing that the learned directions align with meaningful lexical and stylistic cues while also encoding additional representation-level signal.

Overall, our results suggest that AI-generated text detection can be viewed as a *representation-space probing problem*, where machine-generated text is identified by the directions along which it differs from human writing in hidden-state space.

2 Related Work

Supervised detection of AI-generated text. A common approach to AI-generated text detection is to train a classifier on human-written and model-generated corpora. These methods typically use pretrained encoders such as BERT and often achieve strong in-domain performance when train and test distributions are matched (Jawahar et al., 2020; Chen et al., 2024). Variants include boundary-based and topology-aware detectors (Kushnareva et al., 2024, 2021), and methods that improve transfer by removing brittle components of encoder representations (Kuznetsov et al., 2024). A closely related recent method, *ReprGuard* (Chen et al., 2025), also uses hidden-state features from a frozen LM rather than fine-tuning an encoder. Unlike ReprGuard, we build *layer-wise discriminative directions* and represent each text by its alignment with these directions at every layer, which yields a compact layer-diagnostic representation and supports interpretability analyses.

Zero-shot and score-based detectors. Another line of work aims to detect AI-generated text without training a dedicated classifier. Early methods rely on token-level statistics such as likelihood, entropy, rank, log-rank, and likelihood-ratio scores (Su et al., 2023; Wu et al., 2024). More recent methods derive zero-shot criteria from a reference LM, including DetectGPT (Mitchell et al., 2023), Fast-DetectGPT (Bao et al., 2023), DNA-GPT (Yang et al., 2023), and Binoculars (Hans et al., 2024). In contrast, SV-Detect is not zero-shot: it learns a lightweight detector from representation-level features rather than directly from text.

Robustness and evaluation under distribution shift. Recent work has emphasized that strong in-domain results are not sufficient for realistic machine-generated text detection (Tufts et al., 2025; Li et al., 2024; Kushnareva et al., 2024). *DetectRL* (Wu et al., 2024) benchmarks transfer across domains, source LLMs, and attack families, while *MIRAGE* (Fu et al., 2025) focuses on generation, polishing, and rewriting. *PADBen* (Zha et al., 2025) further targets paraphrase-based laundering, iterative paraphrase depth, and human-edited AI-generated text, complementing the attack families covered by *DetectRL* and *MIRAGE* with a laundering-centric perspective. Following this robustness-centered evaluation paradigm, we evaluate SV-Detect on these benchmarks and additionally test cross-benchmark transfer.

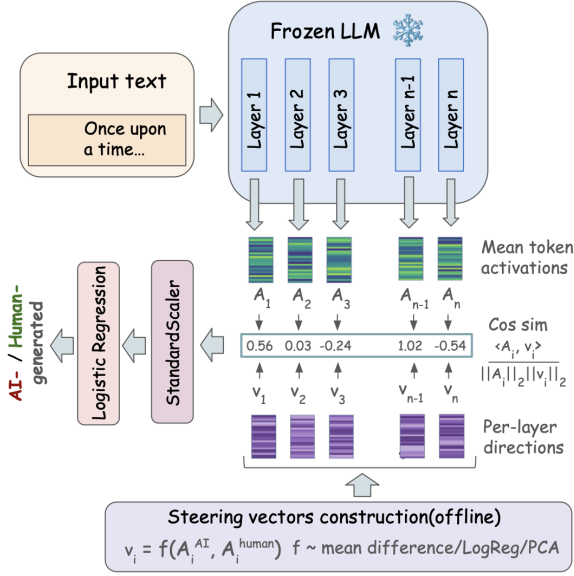


Figure 2: Overview of SV-Detect. A frozen LLM is used to extract mean-pooled hidden activations from each layer. Activations are projected onto layer-wise steering vectors, and the resulting cosine similarity scores are standardized and passed to a logistic regression classifier for AI-generated text detection.

3 Methodology

3.1 Overview

The key idea of our method is based on the hypothesis that human-written and AI-generated texts induce systematically different activation patterns inside a pretrained model. We capture these differences as directions in representation space and use them to construct features for a downstream detection model.

Our pipeline is illustrated in Fig. 2 and consists of four stages: (i) extracting layer-wise activations from a frozen language model, (ii) extracting *AI-generated text directions* that distinguish human-written and machine-generated texts, (iii) projecting text representations onto these directions to obtain low-dimensional features, and (iv) training a lightweight classifier on top of these features.

This design transforms high-dimensional hidden states into interpretable scores that quantify how strongly a text aligns with AI-generated text directions across the network.

3.2 Layer-wise text representations

Let f be a frozen transformer language model with L layers. Given an input text x tokenized into T tokens, let $H_l(x) \in \mathbb{R}^{T \times d}$ denote the hidden states at layer $l \in \{1, \dots, L\}$, where d is the hidden dimension. We construct the text representation at

layer l by mean-pooling hidden states across the token dimension:

$$a_l(x) = \frac{1}{T} \sum_{t=1}^T H_l(x)_t \in \mathbb{R}^d.$$

The full representation of x is therefore the collection of its representations at each layer:

$$a(x) = (a_1(x), \dots, a_L(x)).$$

3.3 Constructing AI-generated text directions

Suppose we are given training data that consists of a set of human-written texts $\mathcal{D}_{\text{real}}$ and a set of AI-generated texts $\mathcal{D}_{\text{fake}}$

$$\mathcal{D}_{\text{real}} = \{x_i^{(r)}\}_{i=1}^{N_r}, \mathcal{D}_{\text{fake}} = \{x_j^{(f)}\}_{j=1}^{N_f}$$

For each layer l , we use the pooled activations

$$a_l^{(r)} = \{a_l(x_i^{(r)})\}_{i=1}^{N_r}, \quad a_l^{(f)} = \{a_l(x_j^{(f)})\}_{j=1}^{N_f}$$

to construct a vector $v_l \in \mathbb{R}^d$ that captures the direction separating AI-generated from human-written text representations. We refer to these directions as *steering vectors*: they are representation-space directions of the same kind used in activation steering and representation engineering work (Zou et al., 2023). However, note that in SV-Detect they are used only as probing and classification features. We study three methods for constructing steering vectors, following common practice in representation engineering (Zou et al., 2023).

Mean-difference. The simplest choice is the normalized difference between class means:

$$\mu_l^{(f)} = \frac{1}{N_f} \sum_{j=1}^{N_f} a_l(x_j^{(f)}), \quad \mu_l^{(r)} = \frac{1}{N_r} \sum_{i=1}^{N_r} a_l(x_i^{(r)}),$$

$$v_l^{\text{mean}} = \frac{\mu_l^{(f)} - \mu_l^{(r)}}{\|\mu_l^{(f)} - \mu_l^{(r)}\|_2}.$$

This vector points from the average human-written representation toward the average AI-generated representation.

Logistic-regression. One alternative is to fit a linear classifier directly in the activation space of each layer and use the normal vector of the separating hyperplane. For each layer l , we train a logistic regression model to separate AI-generated and human-written text activations. Let w_l denote the learned weight vector for layer l . We define the steering vector for this layer as its normalized version:

$$v_l^{\text{logreg}} = \frac{w_l}{\|w_l\|_2}.$$